

Technology Stack

Date	15 July 2026
Team ID	SWTID-2026-3016
Project Name	Credit card approval system
Maximum Marks	2 Marks

Technology Stack Details

<u>S.No</u>	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap 5, JavaScript	Used to build a responsive, user-friendly web interface for collecting applicant details and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, loads the trained Random Forest model, and returns prediction results efficiently.
3	Database / Data Storage	Joblib (.pkl Model File)	The trained Random Forest model is stored as a .pkl file using Joblib, enabling fast loading and prediction without requiring a database.
4	Cloud / Hosting / Deployment	Render	Used to deploy the Flask web application online with minimal configuration and free cloud hosting support.
5	Version Control & CI/CD	GitHub	Used to manage source code, track changes, and integrate directly with Render for automatic deployment from the repository.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn	Scikit-learn is used for training the Random Forest model, Pandas and NumPy for data preprocessing, and Matplotlib/Seaborn for data visualization and model evaluation during development.